NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Course: Electric Circuit I Lab (EEE141L) | **Spring 2026**

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LAB 8 MULTISIM MANUAL

Task:
Construct a RC circuit on Multisim where R = 5 kΩ and C = 0.22 μF. *The input signal is **a square wave** with a **peak-to-peak voltage of 10 V**.*

Then, simulate the circuit and record the following parameters for input signal time periods of 10RC, 30RC, and 3.5 ms:

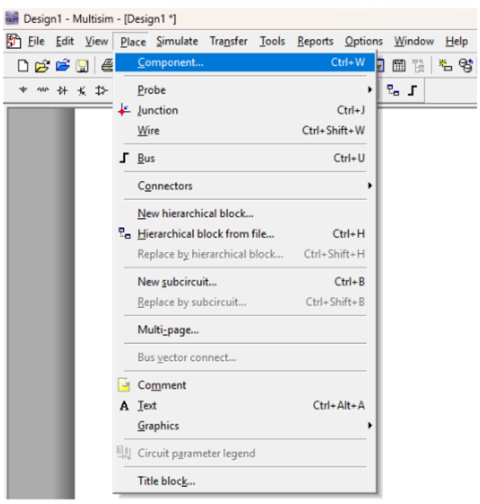
- Frequency of input signal
- Time constant, τ
- Final output, V_C
- Time taken for the capacitor to charge up to V_C
- Time when the capacitor starts to discharge
- Time when the capacitor stops discharging

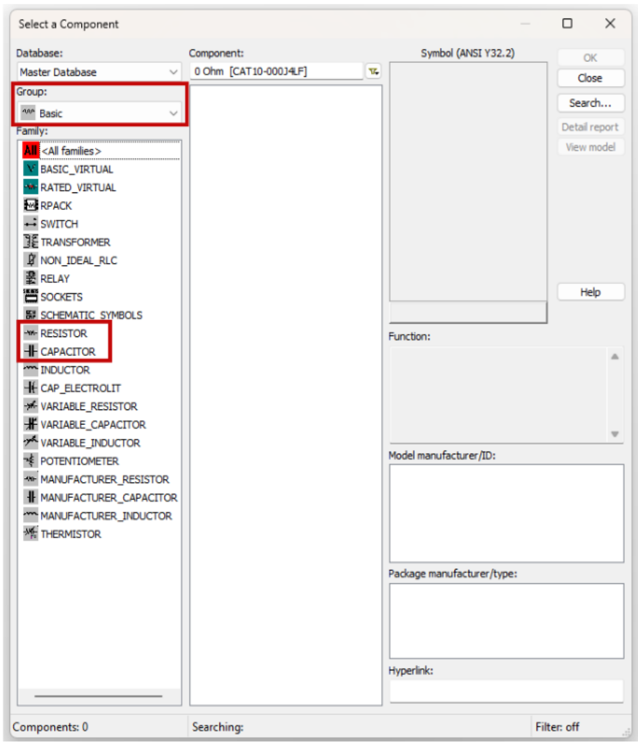
PHASE I: CONSTRUCTING THE RC CIRCUIT

Step 1: Placing the components

The RC circuit consists of two components: a 5 kΩ resistor and a 0.22 μF capacitor. To begin building the circuit in Multisim, go to the **Component...** tab located in the top menu bar. From there, navigate to the **Basic** group to locate the resistor and capacitor. Select each component individually and place them onto the workspace.

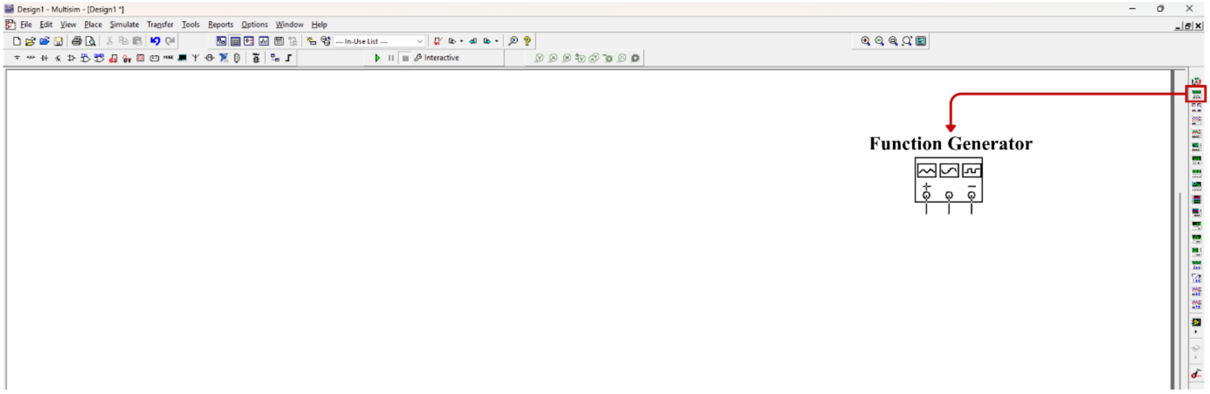
Use the image provided below as reference.





Step 2: Placing the Function Generator

An input signal is required for the circuit, which can be generated using a Function Generator. In Multisim, the Function Generator is available in the **Instruments toolbar** on the right-hand side of the interface, as shown in the image below. Select the Function Generator from this panel and place it onto the workspace to use it as the input source for the circuit.

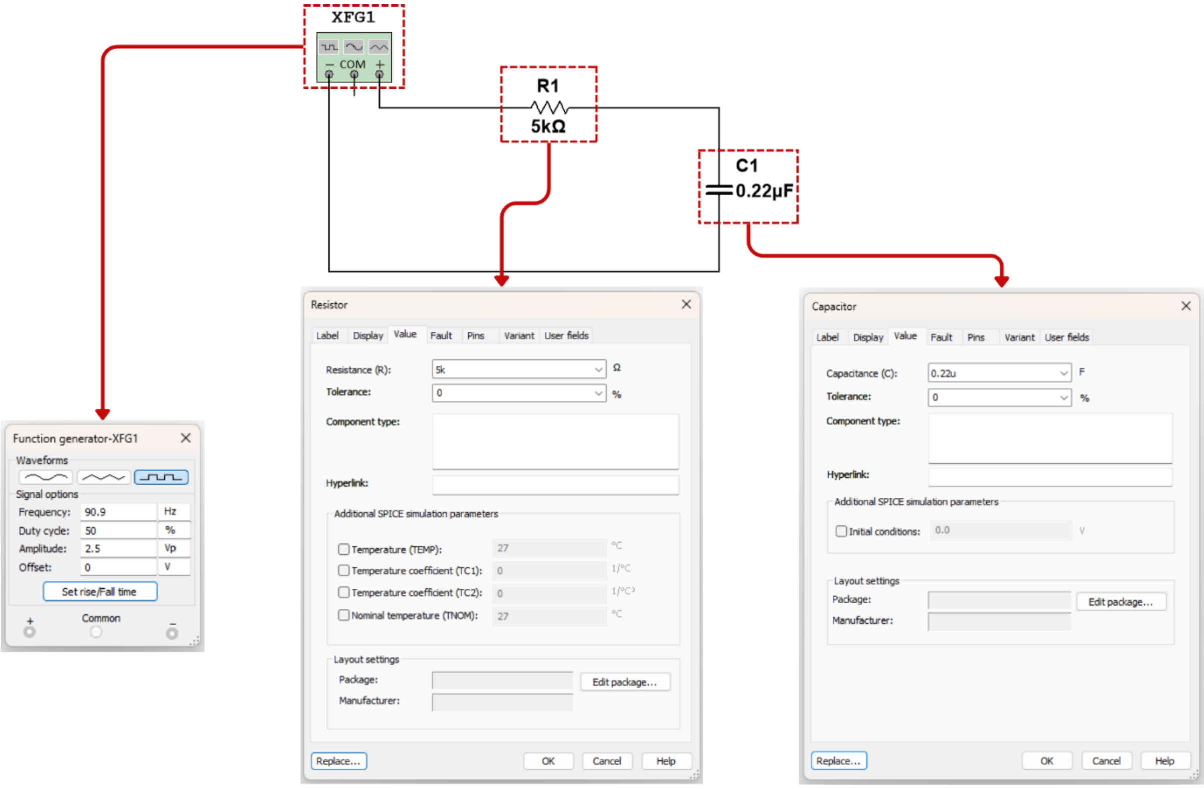


Step 3: Configuring the parameters of the Resistor, Capacitor, and Function Generator

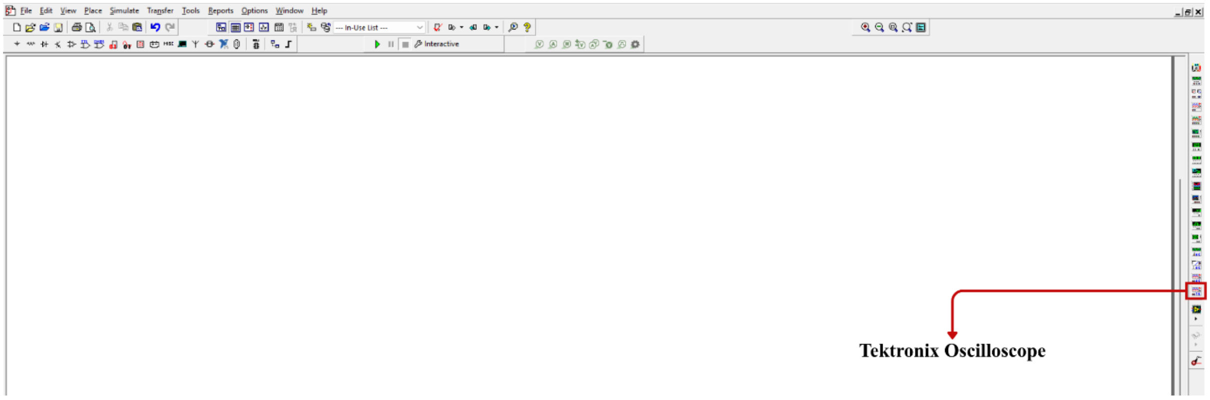
Once the resistor, capacitor, and function generator have been placed on the workspace, the next step is to configure their parameters. Set the resistance of the resistor as "5k" and the capacitance of the capacitor to "0.22u" as required for the circuit.

Next, adjust the settings of the function generator to define the input signal. The frequency should be set according to the given time periods: 90.9 Hz for T = 10RC, 30.3 Hz for T = 30RC, and 285.7 Hz for T = 3.5ms. The input signal has a peak-to-peak voltage of 10 V. In Multisim, this corresponds to setting the amplitude to 2.5 V_p (since $V_{pp} = 4 \times V_p$ for the square wave configuration used).

Make sure all these values are correctly entered before running the simulation. Use the image provided below as reference.

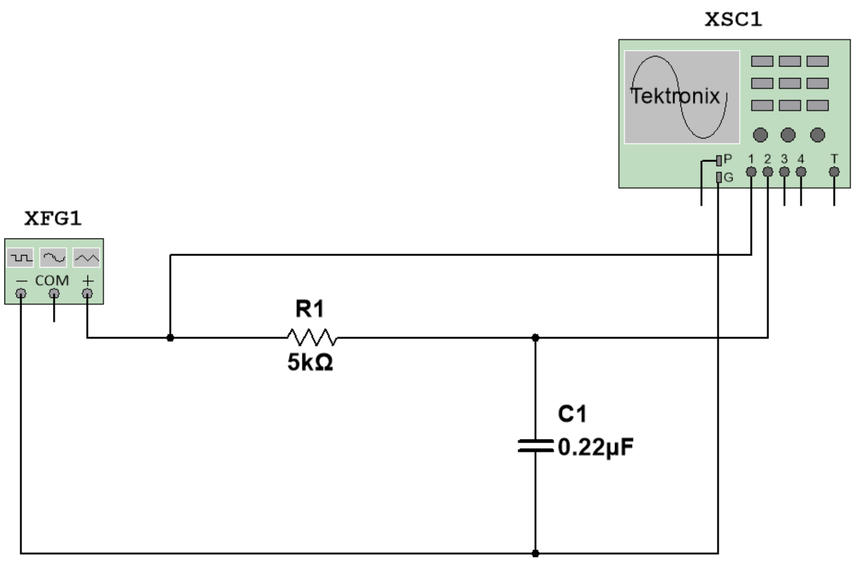


Step 4: Connecting the Tektronix Oscilloscope to the circuit
To obseve both the input from the function generator and the output across the capacitor, an oscilloscope is required. In Multisim, the Tektronix oscilloscope can be found at the bottom of the **Instruments toolbar** on the right-hand side of the interface. Place the oscilloscope onto the workspace.



Next, make the necessary connections. Connect the common ground (G) of the oscilloscope to the negative terminal of the function generator/capacitor. Then, connect Channel 1 to the positive terminal of the function generator to observe the input signal. Finally, connect Channel 2 to the positive terminal of the capacitor to monitor the output voltage across it.

Use the image provided below as reference.

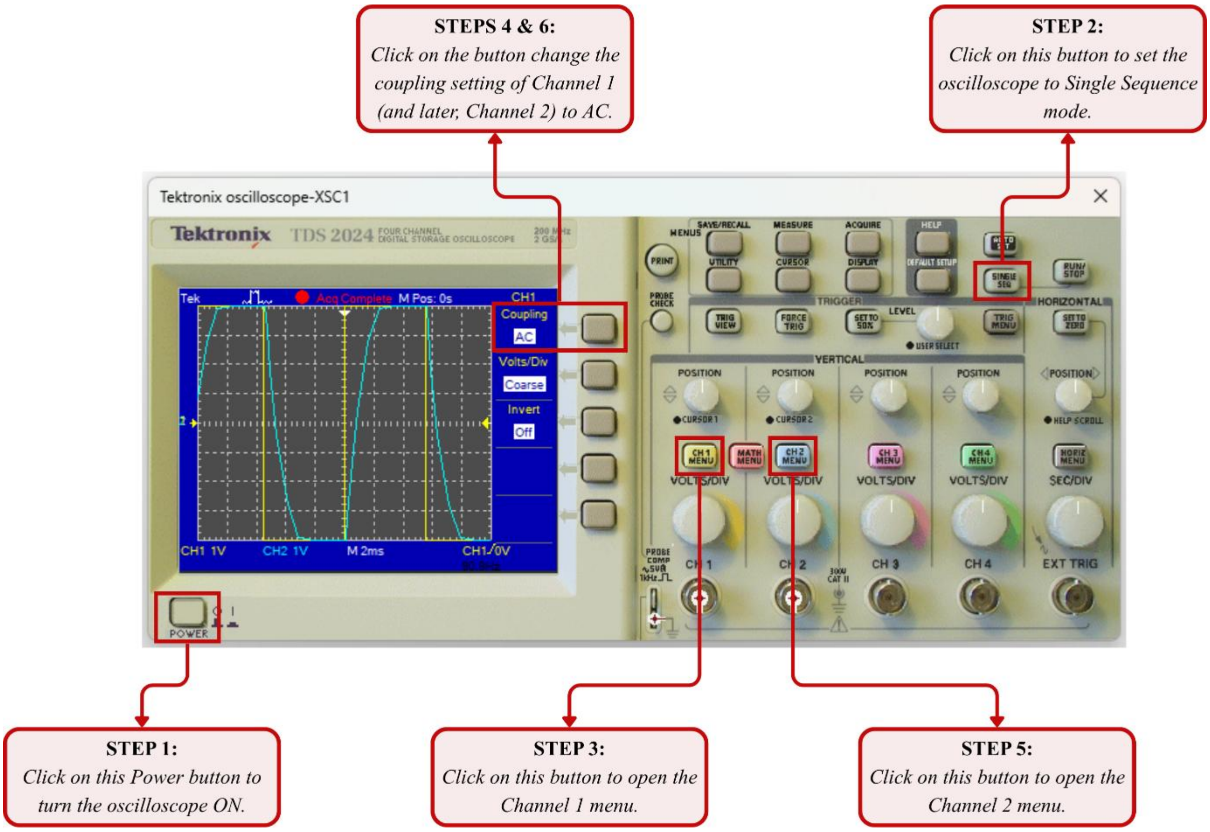


PHASE II: FINDING THE PARAMETERS WHEN T = 10RC

Step 1: Setting the Oscilloscope
After building the circuit and configuring all the required parameters, start the simulation and open the oscilloscope. Turn it on using the **Power** button.

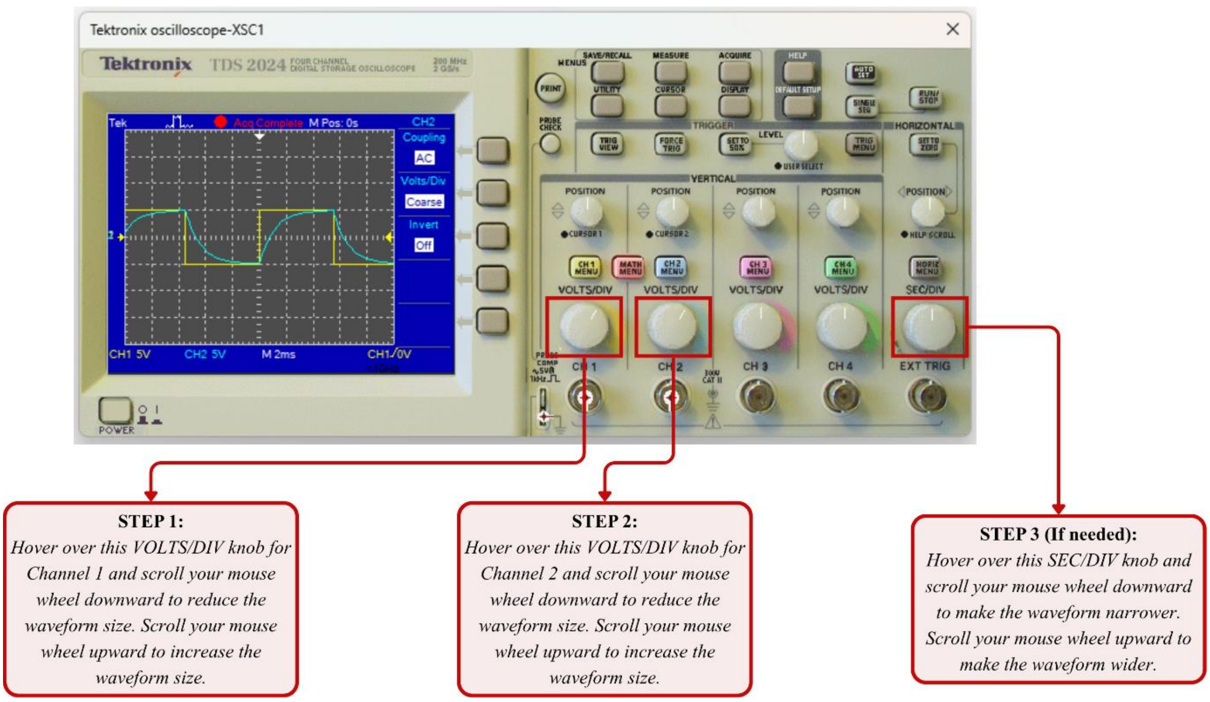
First, set the oscilloscope to **Single Sequence** mode. Then, adjust both Channel 1 and Channel 2 to AC coupling. To do this, open the **CH1 menu** for Channel 1 and the **CH2 menu** for Channel 2, and in each menu, change the coupling setting to AC.

Use the image provided below as reference.



Next, adjust the scale of the signals so that the entire waveforms are clearly visible on the oscilloscope display. This can be done using the **VOLTS/DIV** controls for each channel. Hover over the VOLTS/DIV knob for Channel 1 and scroll your mouse wheel downward to reduce the waveform size (i.e., increase the volts per division). Repeat the same process for Channel 2 by hovering over its VOLTS/DIV knob and scrolling down to appropriately scale the signal.

You can adjust the time scale of the signals using the SEC/DIV knob on the side labelled “Horizontal”. By modifying this control, you can make the waveforms appear wider or narrower on the display.



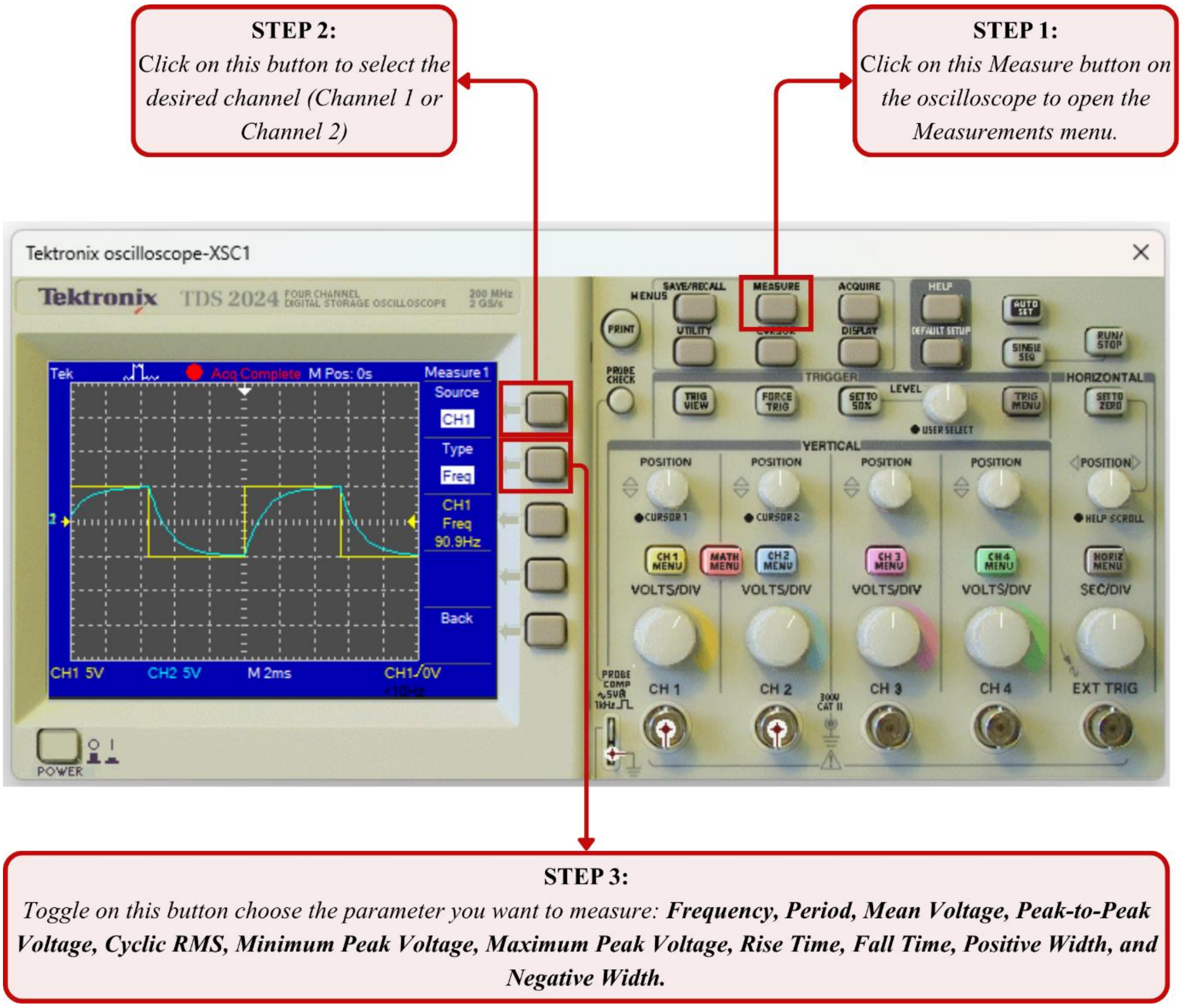
Step 2: Measuring frequency and peak-to-peak voltage of each signal

To measure the frequency and peak-to-peak voltage of the signals on each channel, first click the **Measure** button on the oscilloscope. This will open the **Measurements** menu.

Begin by selecting the desired channel (Channel 1 or Channel 2) using the button located at the top beside the oscilloscope display. Once the channel is selected, choose the parameter you want to measure by toggling the button just below the channel selection button.

The measurement menu provides a wide range of parameters, including: ***Frequency***, ***Period***, Mean Voltage, ***Peak-to-Peak Voltage***, Cyclic RMS, Minimum Peak Voltage, Maximum Peak Voltage, Rise Time, Fall Time, Positive Width, and Negative Width. **The final output voltage, V_C, is taken as the peak-to-peak voltage measured from the Channel 2 signal on the oscilloscope.**

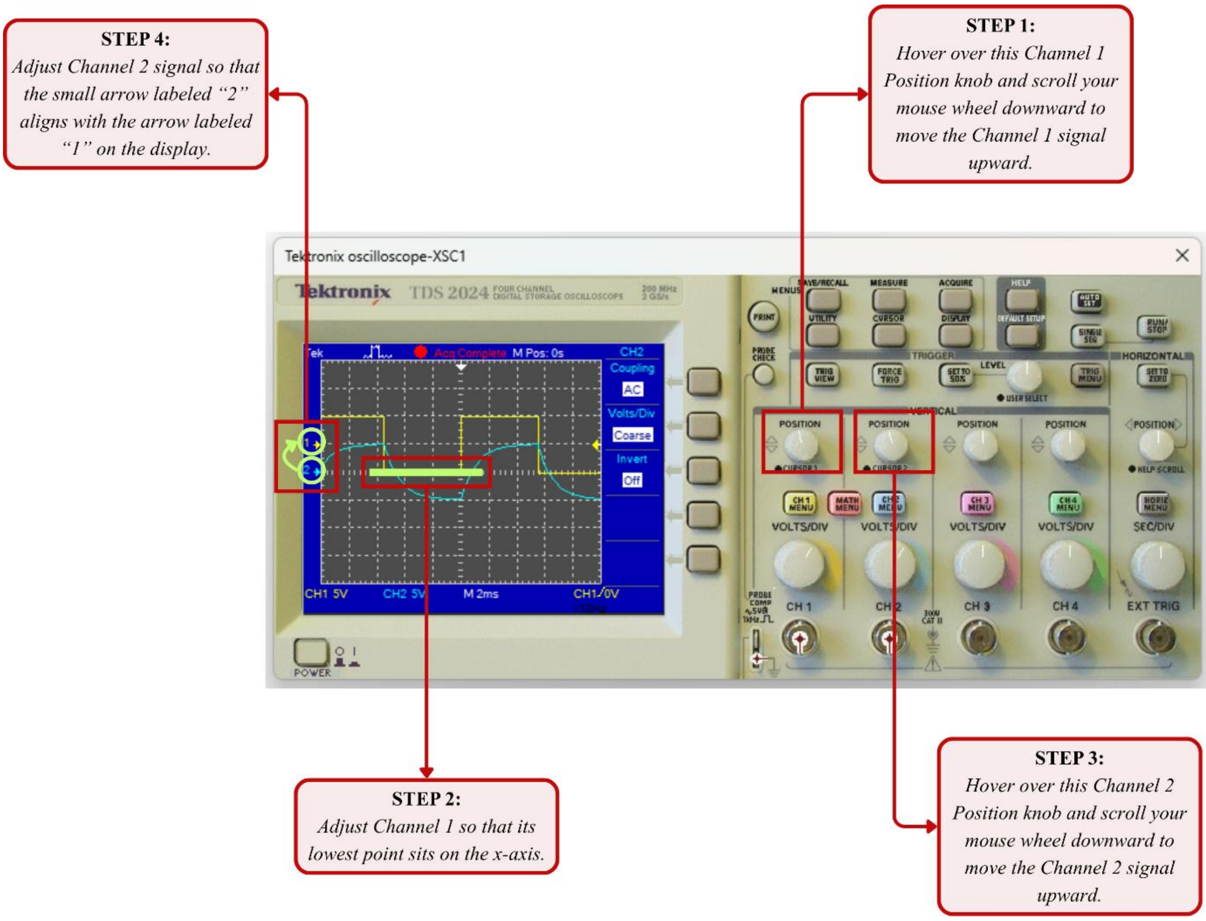
Use the image provided below as reference.



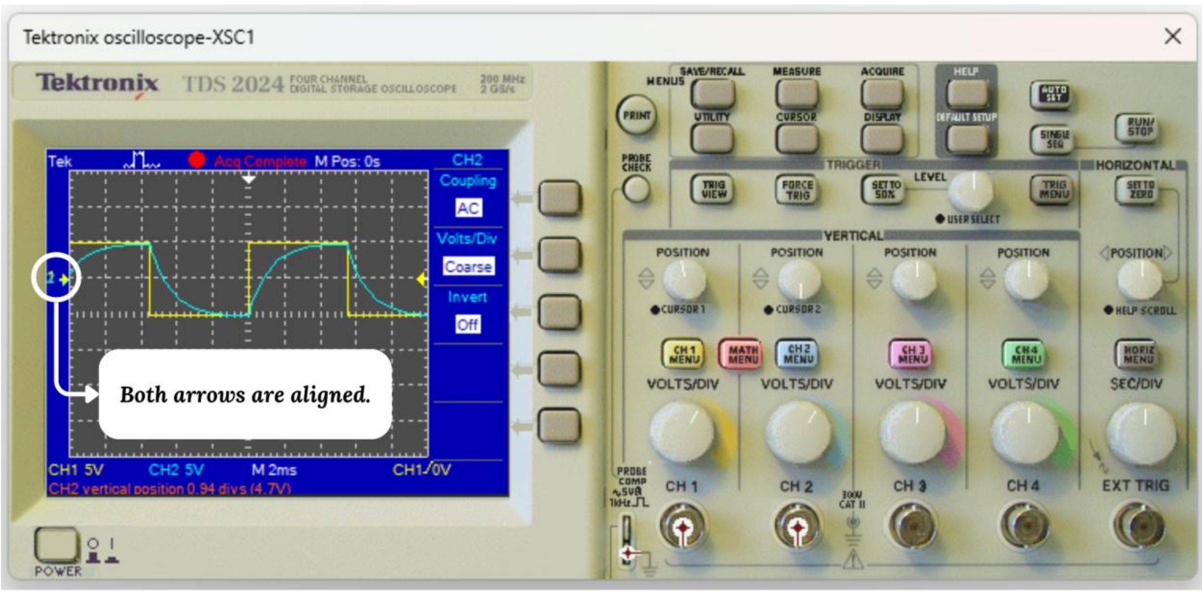
Step 3: Measuring the time constant τ

To measure the time constant from the oscilloscope, begin by adjusting the vertical positions of both signals. First, shift the waveforms upward so that the minimum peak of the Channel 1 signal aligns with the x-axis. This can be done using the **Position** knobs for each channel—hover over the knob and scroll your mouse wheel downward to move the signal upward.

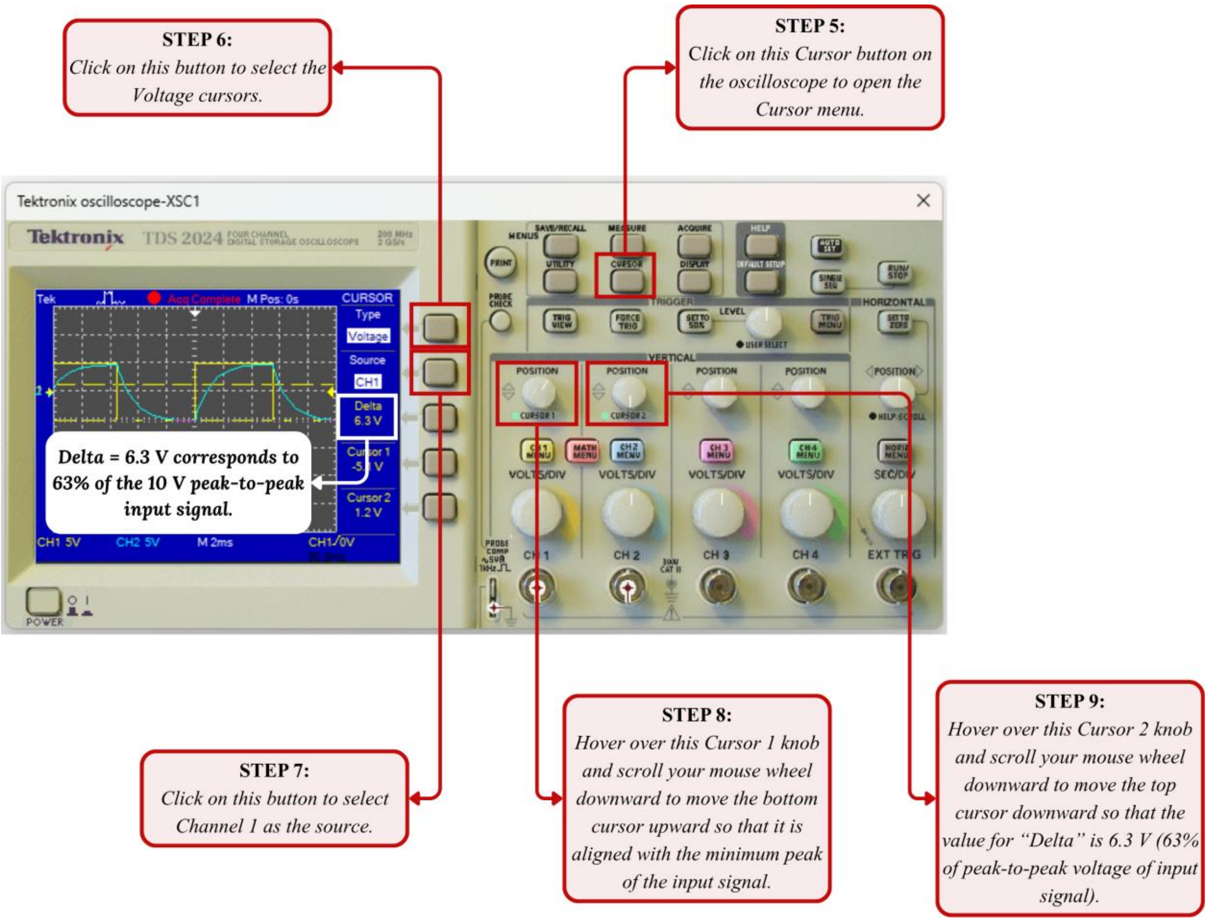
Start by adjusting Channel 1 so that its lowest point sits on the x-axis. Then, adjust Channel 2 so that its midpoint aligns with the midpoint of the Channel 1 signal. For better accuracy, align the small arrow labeled “2” with the arrow labeled “1” on the display.



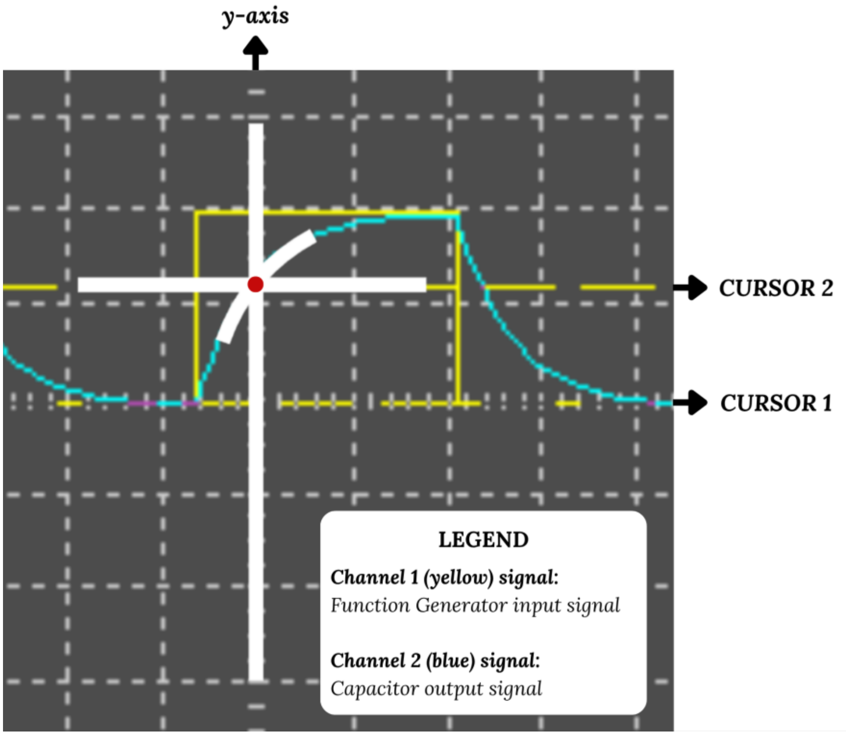
Note: *When T ≥ 10RC, you can directly align the Channel 2 signal with the x-axis instead of following the full alignment process. However, for T < 10RC, the detailed alignment method described above must be followed. To avoid confusion and ensure consistency, it is recommended to use a single standard procedure for all cases.*



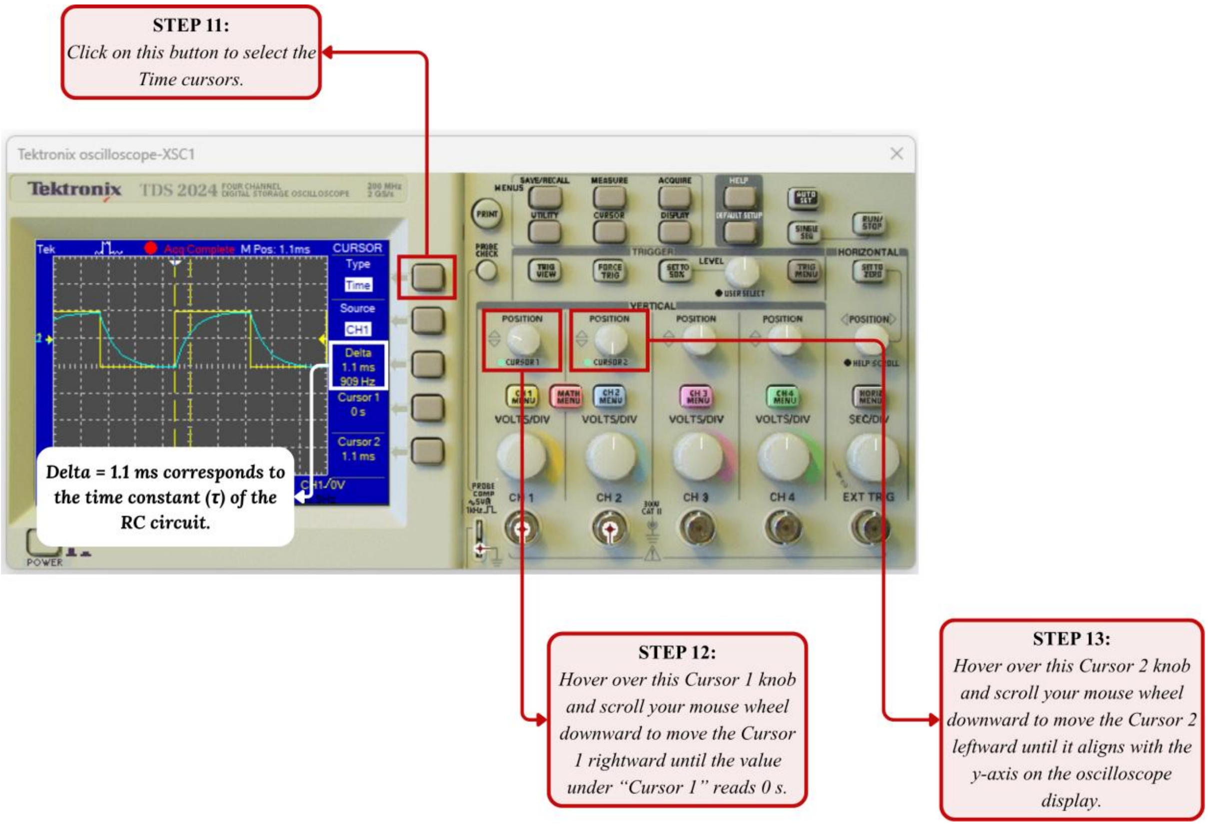
After properly aligning both signals, use the cursor feature to determine the point on the Channel 2 waveform where the capacitor reaches 63% of its final value. Start by clicking the **Cursor** button on the oscilloscope to open the cursor menu. Then, use the button located at the top beside the display to select the Voltage cursor type. Make sure the source is set to CH1 by toggling the button just below the cursor type selection. Next, adjust the cursors. Using the Cursor 1 control knob, move Cursor 1 upward until it aligns with the minimum peak of the Channel 1 signal (scroll the mouse wheel downward to move the cursor up). Then, use the Cursor 2 control knob to position Cursor 2 at a point where the "Delta" value shown on the screen is 6.3 V, which corresponds to 63% of the 10 V peak-to-peak input signal.



Next, hover over the Horizontal Position knob and use your mouse scroll wheel to shift both waveforms horizontally. Adjust them until the point where the Channel 2 signal intersects Cursor 2 is aligned with the y-axis on the oscilloscope display.



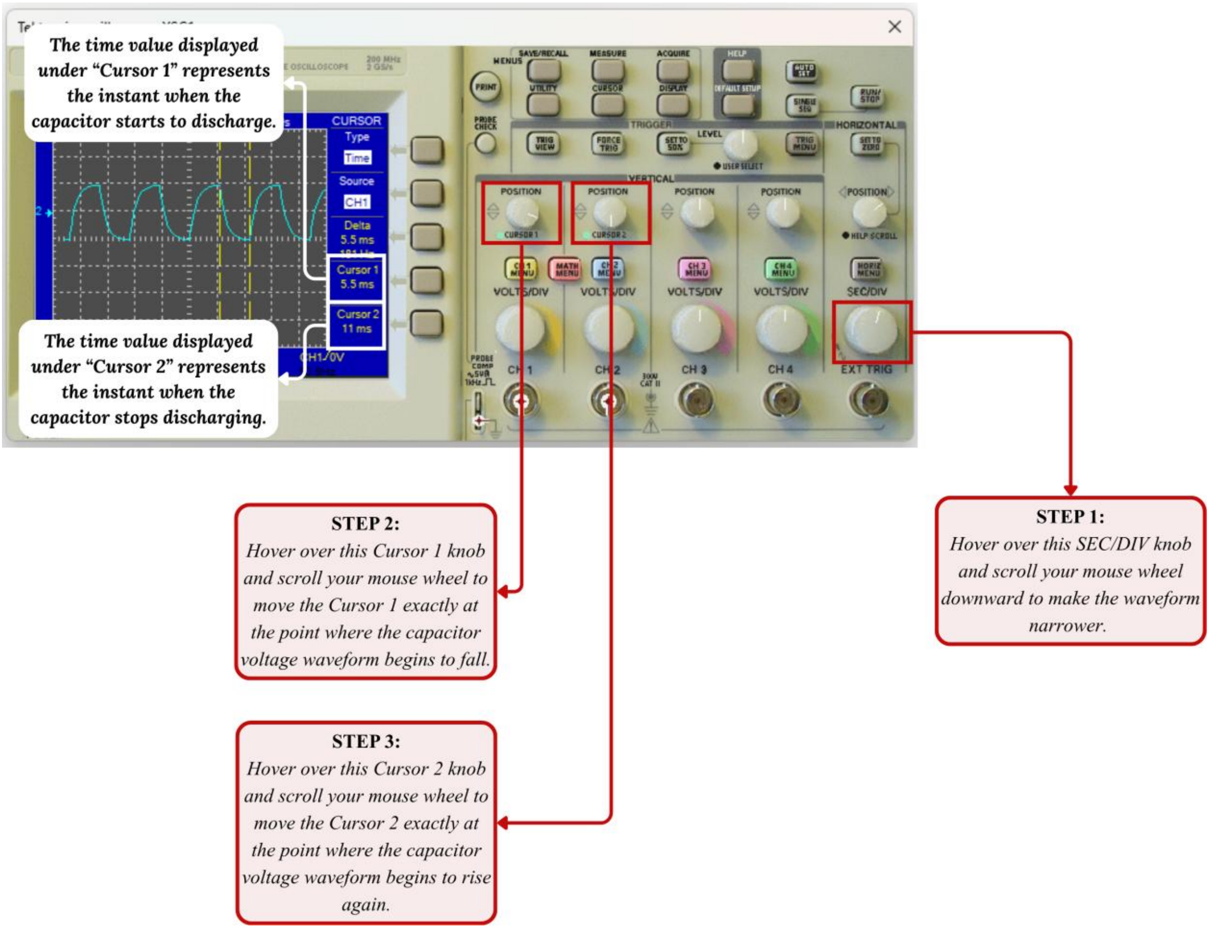
Next, click the button at the top beside the display again and switch to the Time cursor type. Ensure that the source remains set to CH1. Now, adjust the cursors accordingly. Using the Cursor 1 control knob, move Cursor 1 to the right until the value under "Cursor 1" reads 0 s (scroll the mouse wheel downward to shift the cursor right). Then, use the Cursor 2 control knob to move Cursor 2 leftward until it aligns with the y-axis on the oscilloscope display. The value shown as "Delta" now represents the time taken for the capacitor to charge up to 63% of its final value. Therefore, this "Delta" value corresponds to the time constant (τ) of the RC circuit.



Step 4: Measuring the time at which the capacitor begins to discharge and the time at which it completes its discharge

To determine the time at which the capacitor starts discharging, use the Cursor 1 control knob and position Cursor 1 exactly at the point where the capacitor voltage waveform begins to fall. The time value displayed under "Cursor 1" represents the instant when the capacitor starts to discharge.

Similarly, to find the time at which the capacitor stops discharging, use the Cursor 2 control knob and place Cursor 2 at the point where the voltage waveform begins to rise again. The time shown under "Cursor 2" indicates the moment when the capacitor completes its discharge and starts charging once more.

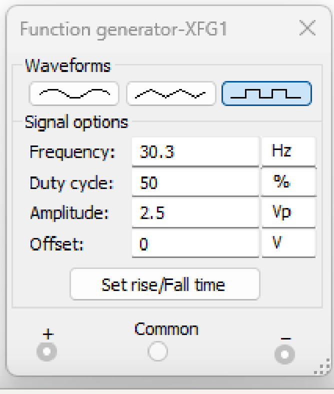


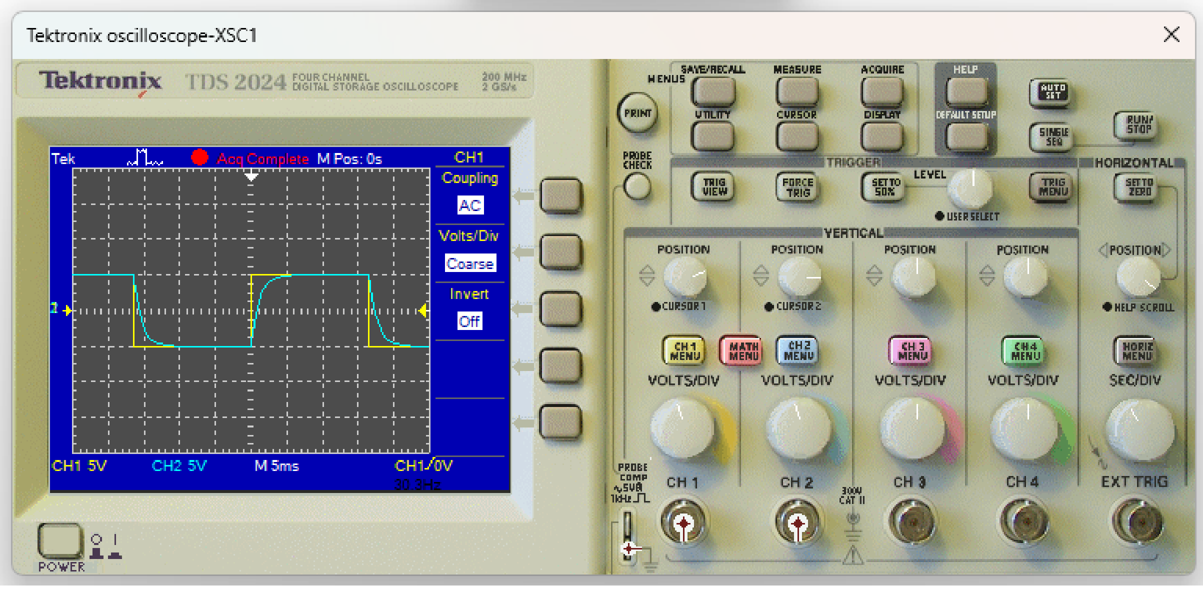
Step 5: Fill up the data table

Measurement	Time period, T = 10RC
Frequency of input signal	**90.9 Hz**
Time constant, τ	**1.1 ms**
Final output, V_C	**10 V_{pp}**
Time taken for the capacitor to charge up to V_C	$5 \times \tau = 5 \times 1.1$ ms = **5.5 ms**
Time when the capacitor starts to discharge	**5.5 ms**
Time when the capacitor stops discharging	**11.0 ms**

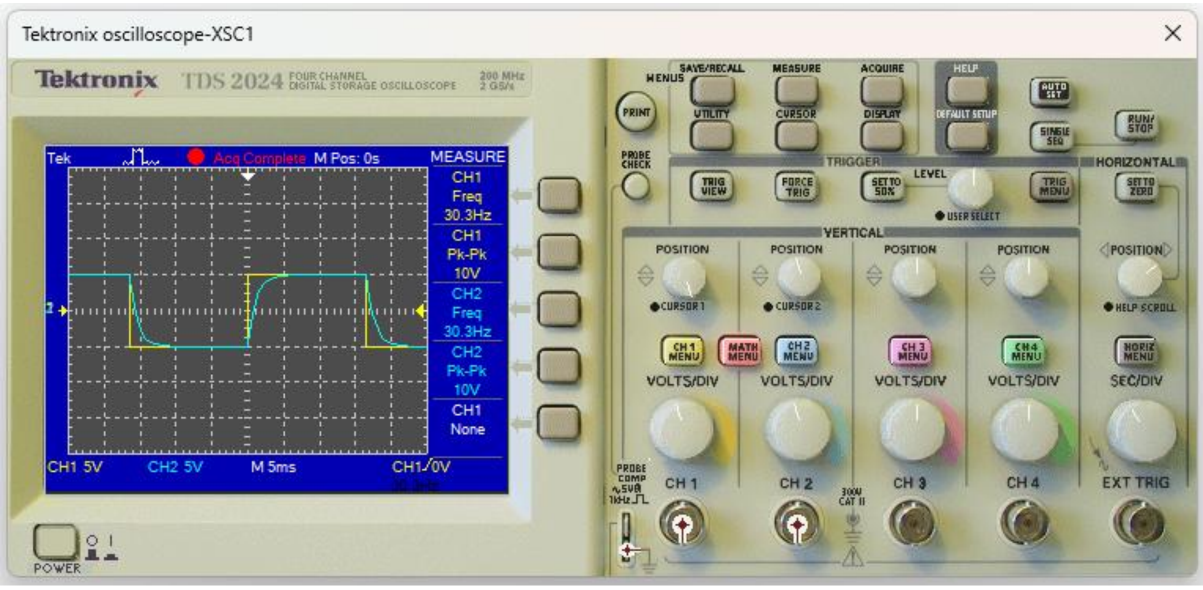
PHASE III: FINDING THE PARAMETERS WHEN T = 30RC

Step 1: Setting the Oscilloscope

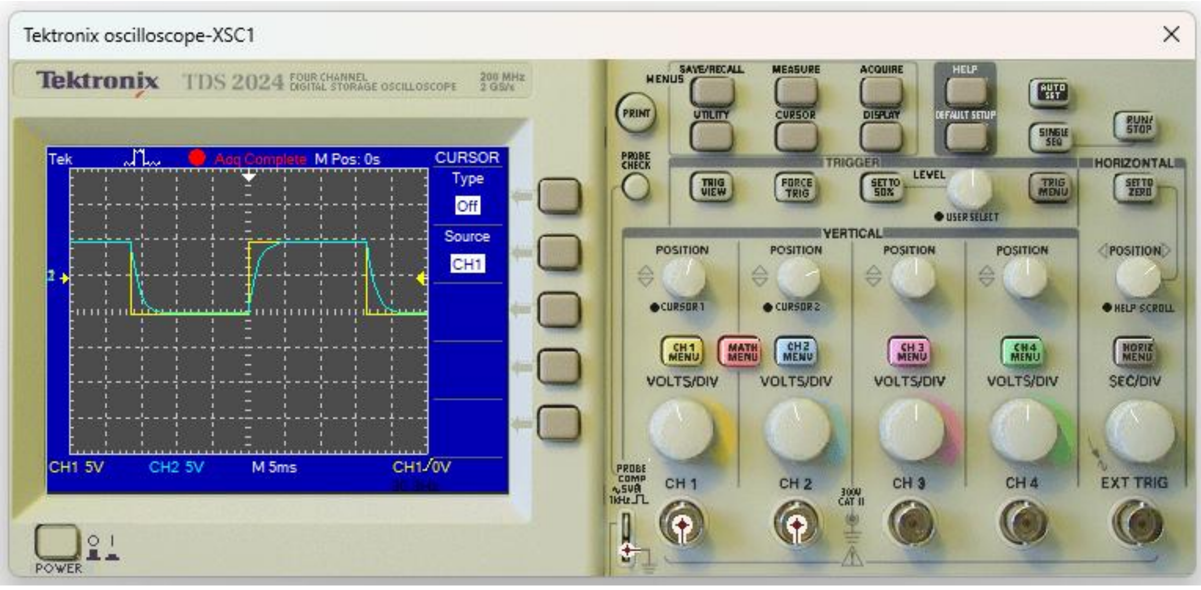


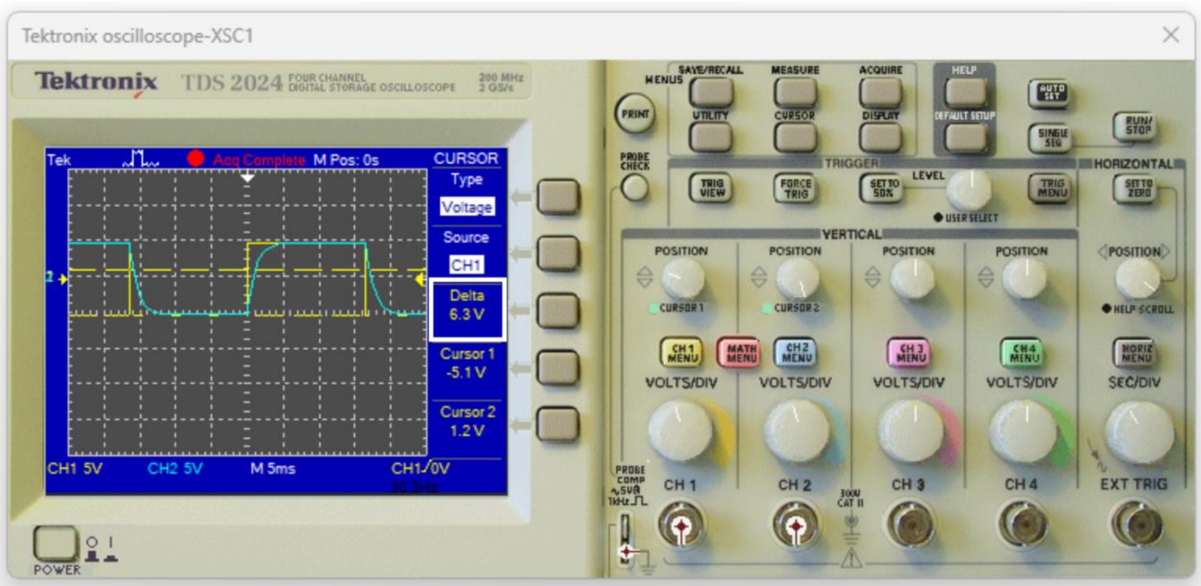


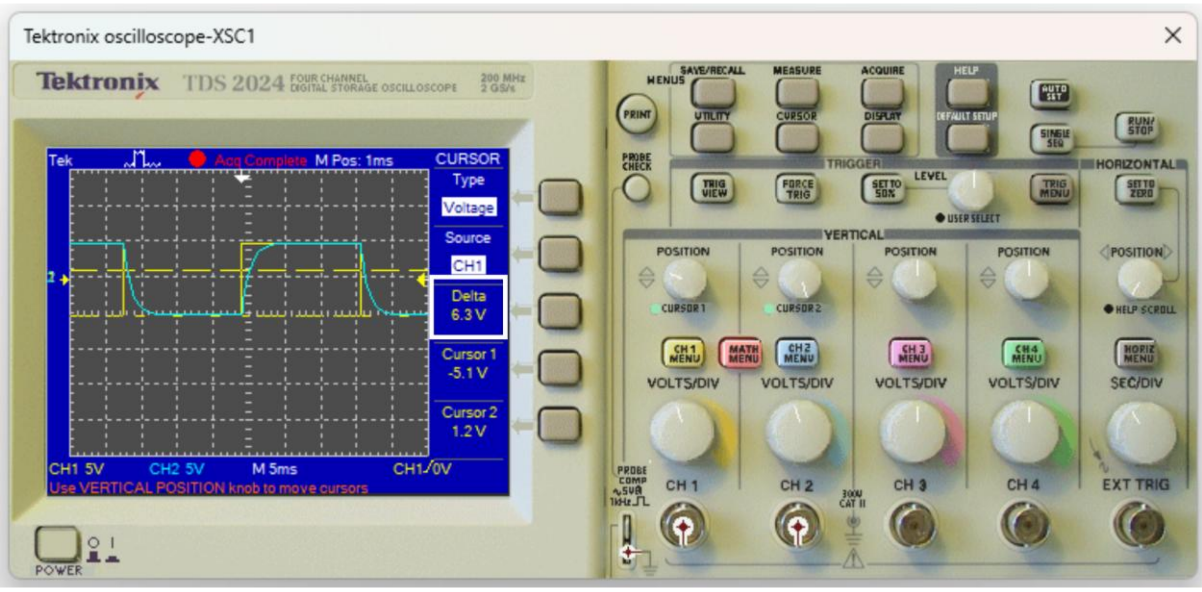
Step 2: Measuring frequency and peak-to-peak voltage of each signal

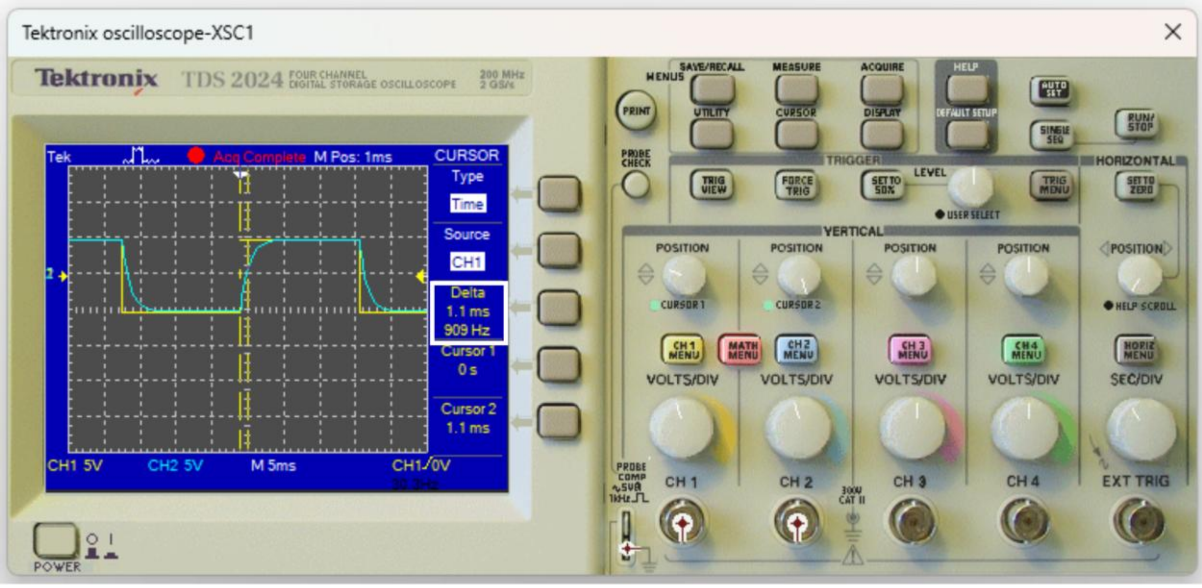


Step 3: Measuring the time constant τ

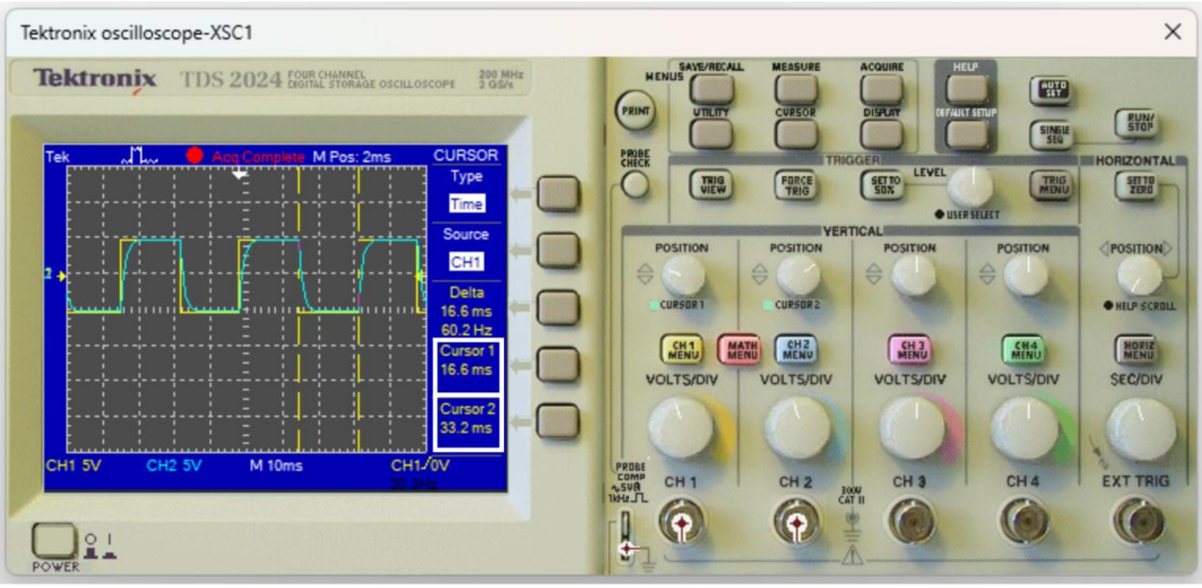








Step 4: Measuring the time at which the capacitor begins to discharge and the time at which it completes its discharge

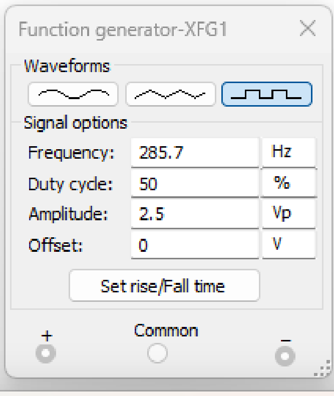


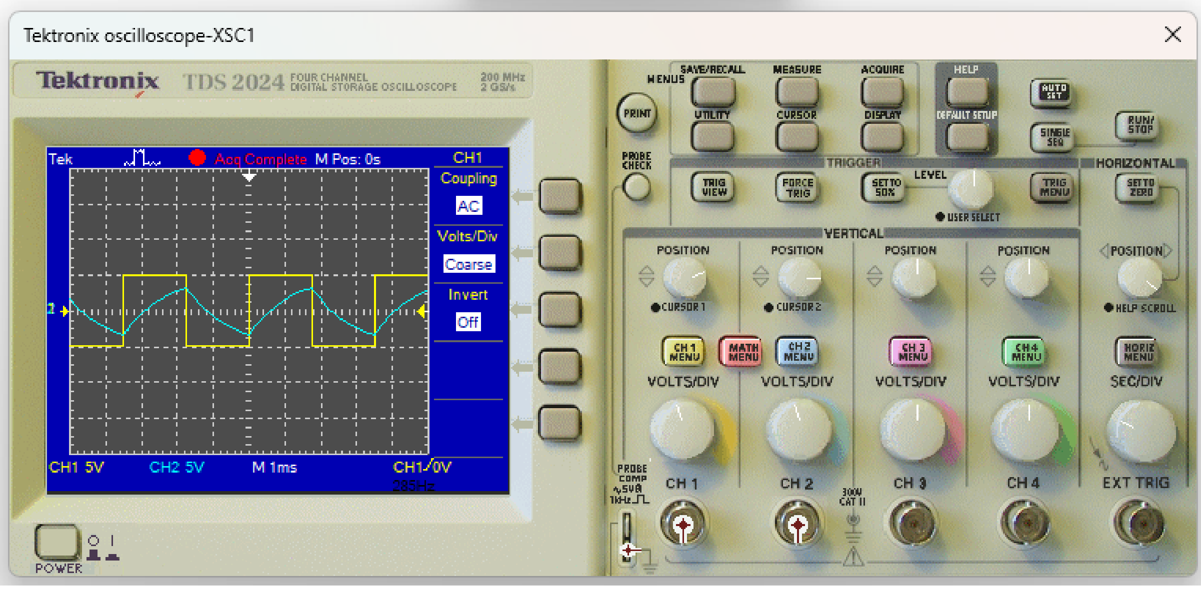
Step 5: Fill up the data table

Measurement	Time period, T = 30RC
Frequency of input signal	**30.3 Hz**
Time constant, τ	**1.1 ms**
Final output, V_C	**10 V_{pp}**
Time taken for the capacitor to charge up to V_C	$5 \times \tau = 5 \times 1.1$ ms = **5.5 ms**
Time when the capacitor starts to discharge	**16.6 ms**
Time when the capacitor stops discharging	**33.2 ms**

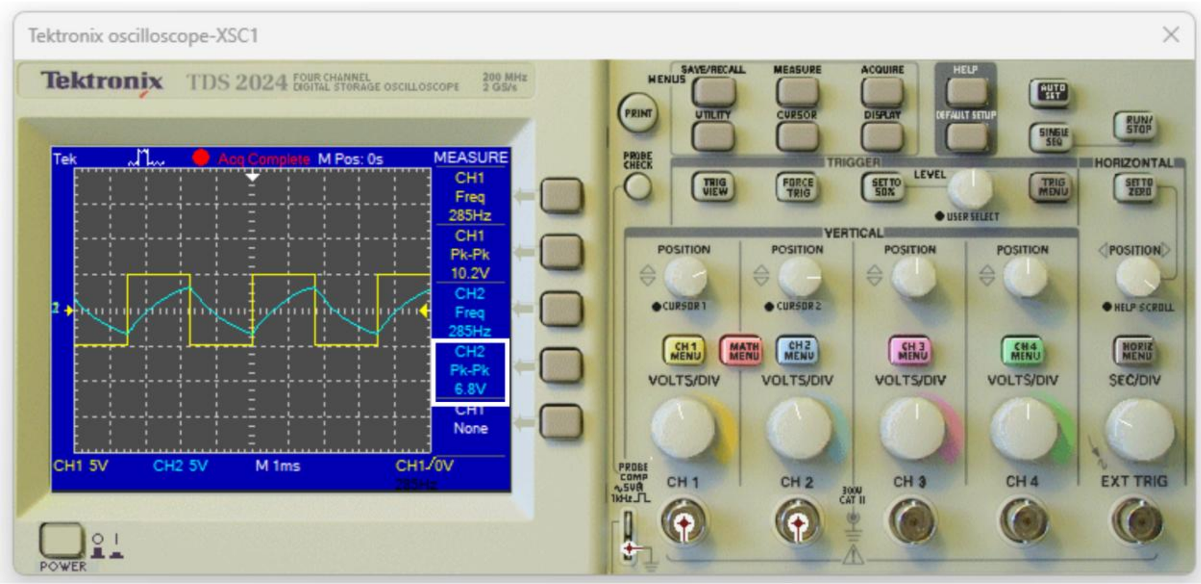
PHASE IV: FINDING THE PARAMETERS WHEN T = 3.5 ms

Step 1: Setting the Oscilloscope

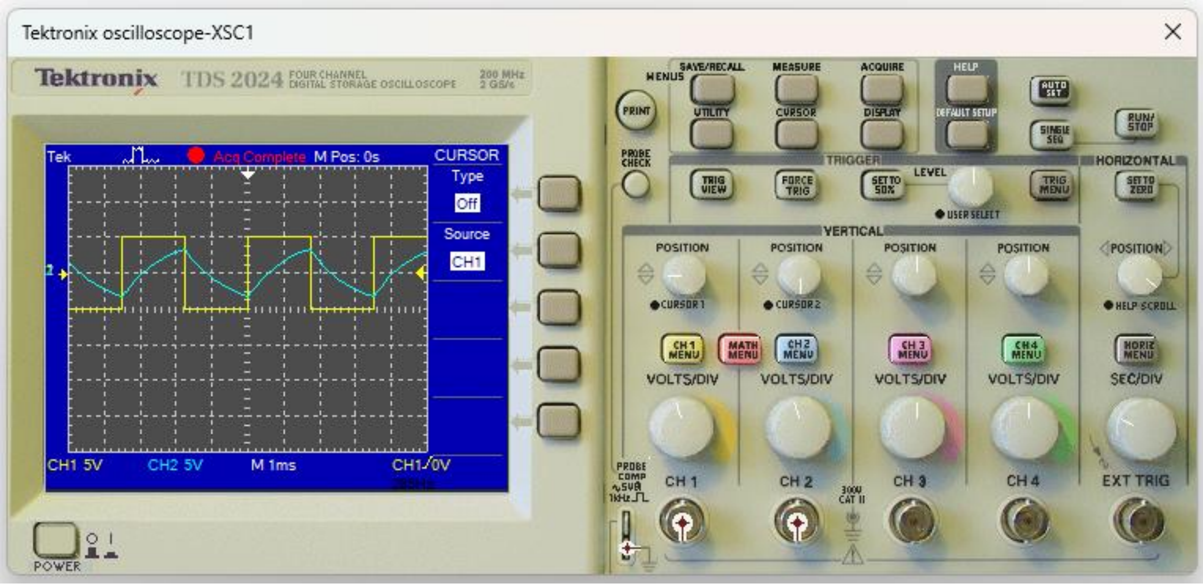


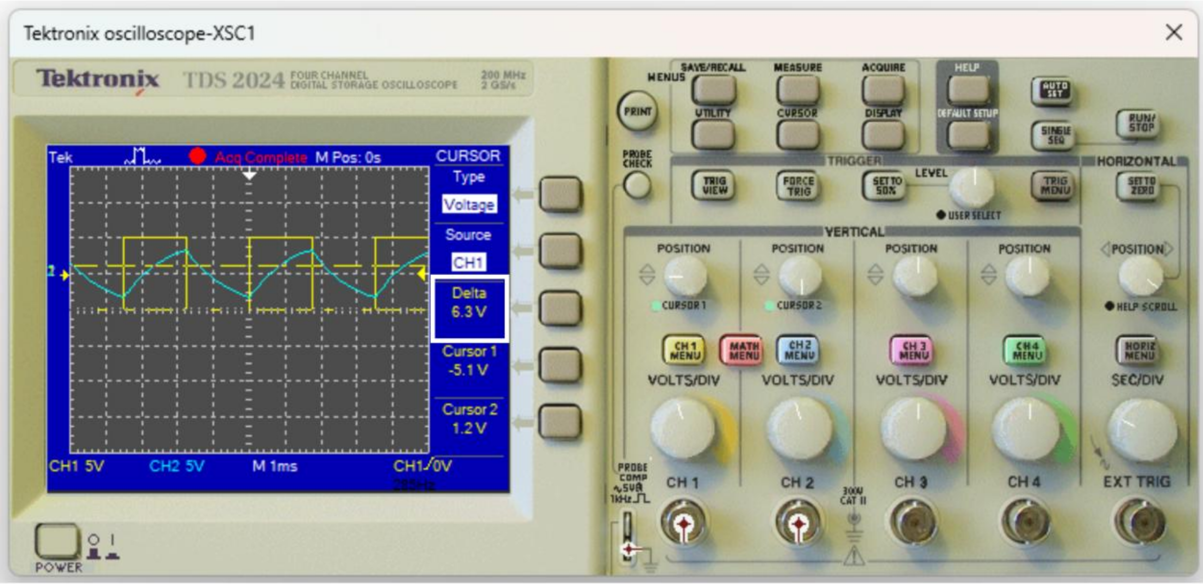


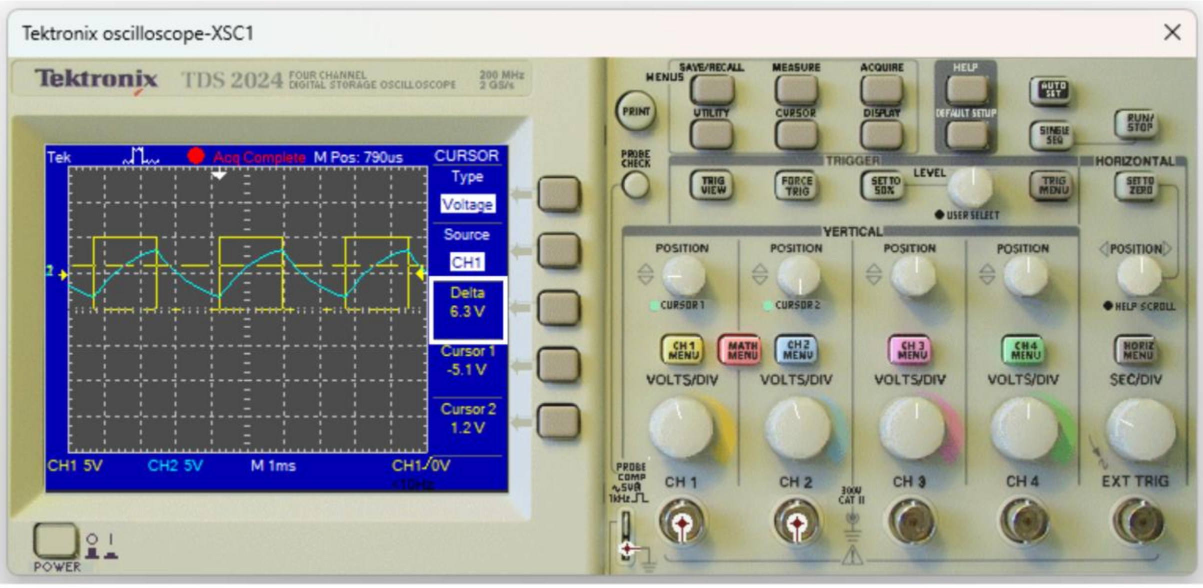
Step 2: Measuring frequency and peak-to-peak voltage of each signal

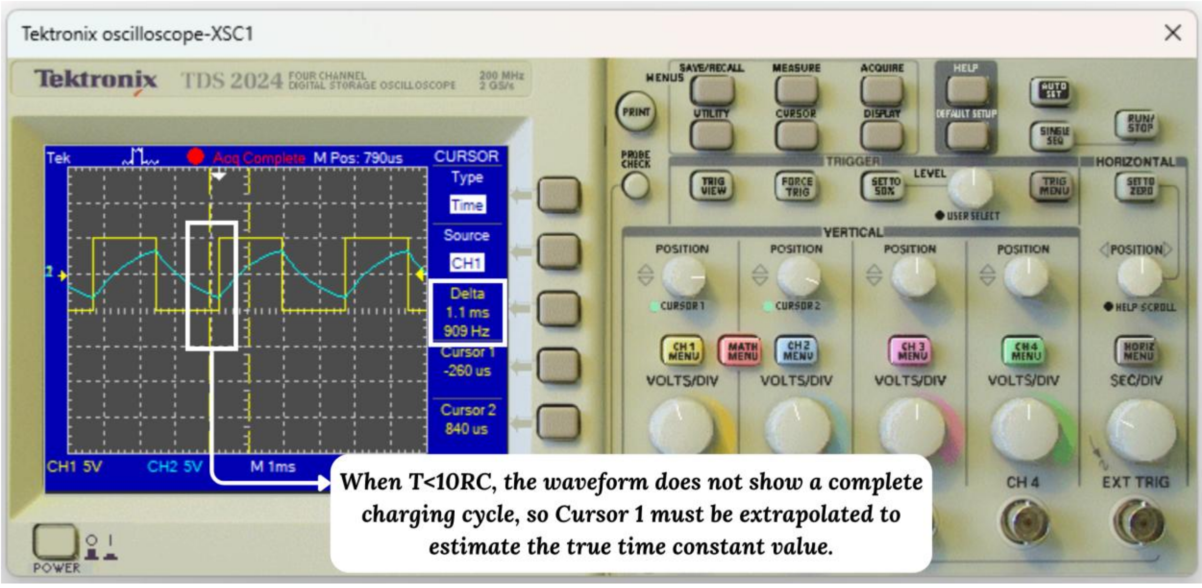


Step 3: Measuring the time constant τ

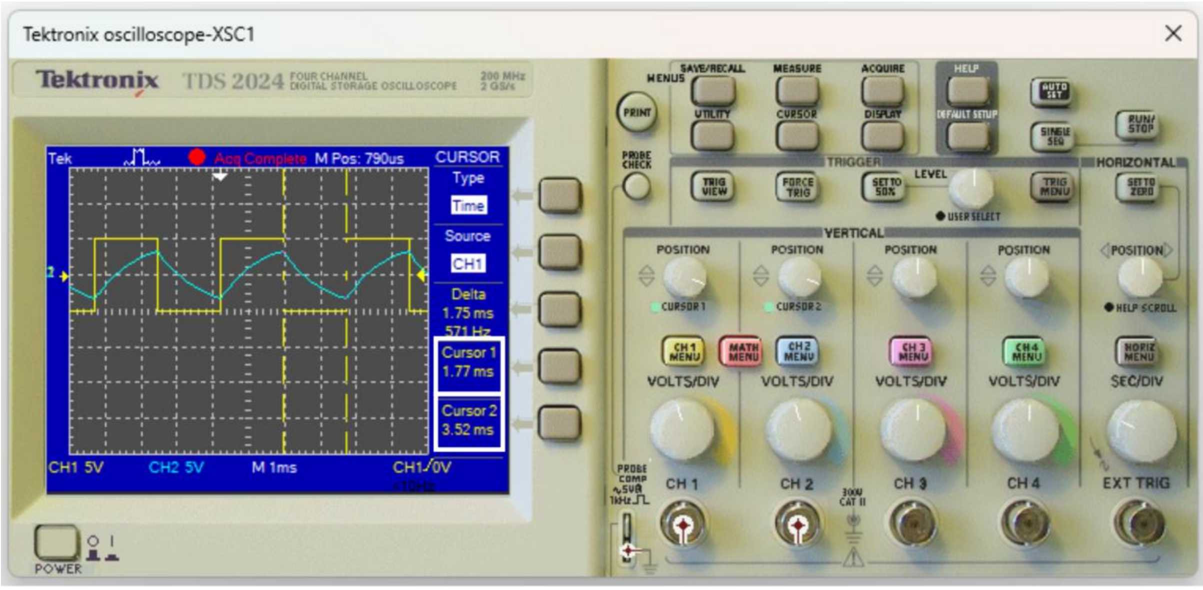








Step 4: Measuring the time at which the capacitor begins to discharge and the time at which it completes its discharge



Step 5: Fill up the data table

Measurement	Time period, T = 30RC
Frequency of input signal	**285.7 Hz**
Time constant, τ	**1.1 ms**
Final output, V_C	**6.8 V_{pp}**
Time taken for the capacitor to charge up to V_C	**1.77 ms** *(Not fully charged)*
Time when the capacitor starts to discharge	**1.77 ms**
Time when the capacitor stops discharging	**3.52 ms**